

Experiment 2

Code :

```
import numpy as np

import pandas as pd

import matplotlib.pyplot as plt

data = pd.read_csv("Salary_Data.csv")

data.columns = data.columns.str.strip()

print("Columns in dataset:", data.columns.tolist())

if "Experience" not in data.columns or "Salary" not in data.columns:

    print("\nError: Required columns 'Experience' and 'Salary' were not found.")

    print("Available columns are:", data.columns.tolist())

    exit()

experience = data["Experience"].to_numpy()

salary = data["Salary"].to_numpy()

plt.figure(figsize=(8,6))

plt.scatter(experience, salary, color='blue', s=70, label='Original Data')

plt.xlabel("Experience (Years)")

plt.ylabel("Salary")

plt.title("Experience vs Salary")

plt.grid(True)

plt.legend()

plt.show()

mean_x = np.mean(experience)

mean_y = np.mean(salary)

b1 = np.sum((experience - mean_x) * (salary - mean_y)) / np.sum((experience - mean_x) ** 2)

b0 = mean_y - b1 * mean_x

print("\nRegression Coefficients")

print("-----")

print("Intercept (b0):", round(b0, 4))

print("Slope (b1):", round(b1, 4))

print("\nRegression Equation")

print("-----")

print(f"Salary = {b0:.4f} + ({b1:.4f}) × Experience")

predicted = b0 + b1 * experience

sorted_index = np.argsort(experience)

experience_sorted = experience[sorted_index]

predicted_sorted = predicted[sorted_index]
```

```
plt.figure(figsize=(8,6))

plt.scatter(experience, salary, color='blue', s=70, label='Original Data')

plt.plot(experience_sorted, predicted_sorted, color='red', linewidth=2, label='Regression Line')

plt.xlabel("Experience (Years)")

plt.ylabel("Salary")

plt.title("Linear Regression: Experience vs Salary")

plt.grid(True)

plt.legend()

plt.show()
```

```
# Interpretation
```

```
print("\nInterpretation")
```

```
print("-----")
```

```
if b1 > 0:
```

```
    print("The slope is positive.")
```

```
    print("As experience increases, salary tends to increase.")
```

```
elif b1 < 0:
```

```
    print("The slope is negative.")
```

```
    print("As experience increases, salary tends to decrease.")
```

```
else:
```

```
    print("No linear relationship exists between experience and salary.")
```

```
# Prediction for a new experience value
```

```
new_experience = 3.5
```

```
predicted_salary = b0 + b1 * new_experience
```

```
print("\nPrediction")
```

```
print("-----")
```

```
print(f"Predicted Salary for {new_experience} years of experience = {predicted_salary:.2f}")
```

Output :

```
Columns in dataset: ['Experience', 'Salary']
```

```
Regression Coefficients
```

```
-----
```

```
Intercept (b0): 83235.9858
```

```
Slope (b1): -22.1931
```

```
Regression Equation
```

```
-----
```

```
Salary = 83235.9858 + (-22.1931) × Experience
```

Interpretation

The slope is negative.

As experience increases, salary tends to decrease.

Prediction

Predicted Salary for 3.5 years of experience = 83158.31

